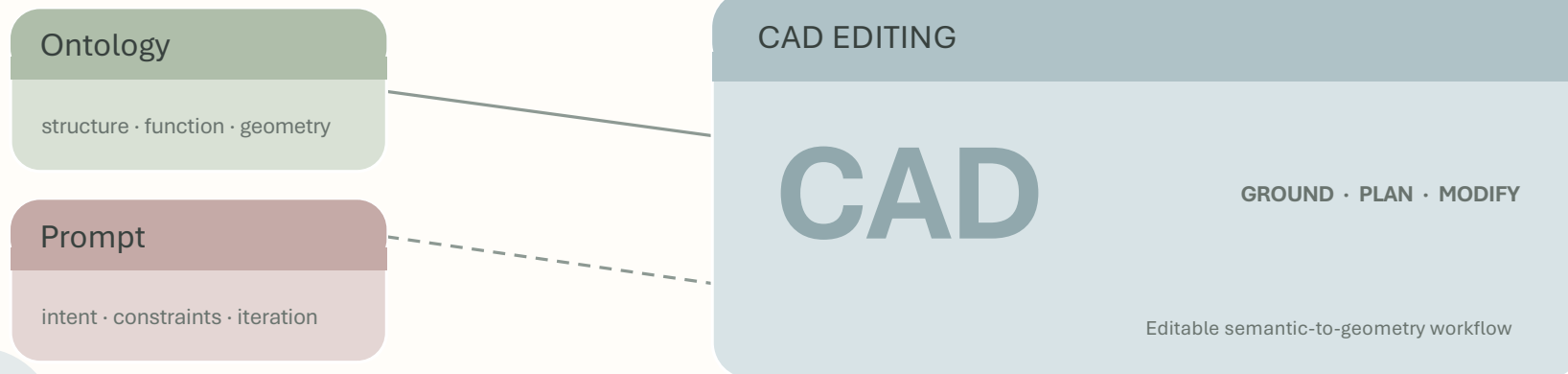


Ontology & Prompt Guided CAD Editing

Semantic grounding and structured prompting for reliable model modification



Underspecified requests raise two research questions

One request may support several interpretations—and several feasible designs.

RESEARCH QUESTION 1

REQUEST AMBIGUITY

How can a CAD agent identify and resolve ambiguity in Function, Reference Geometry, and Operation?

Interpret intent without over-assuming.

RESEARCH QUESTION 2

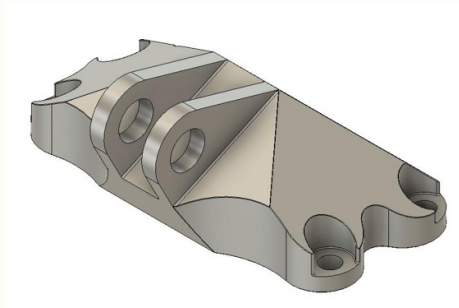
DESIGN FEASIBILITY

How can a CAD agent generate diverse yet feasible designs while preserving the request's functional intent?

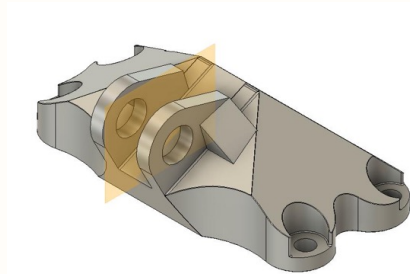
Explore alternatives without violating constraints.

REQUEST AMBIGUITY & DESIGN FEASIBILITY EXAMPLE

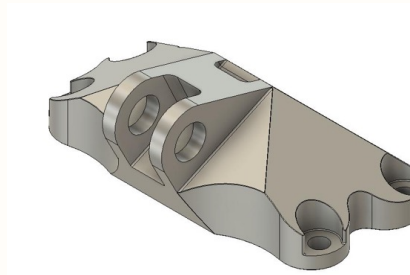
Add a 1.5-millimetre rib to increase support and make the structure strong.



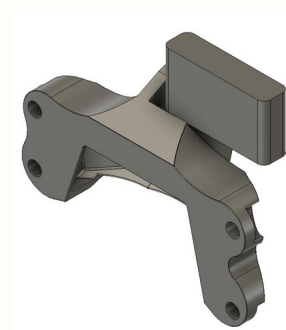
Designer 1



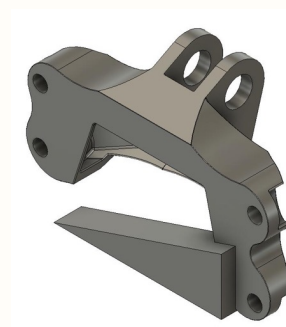
Designer 2



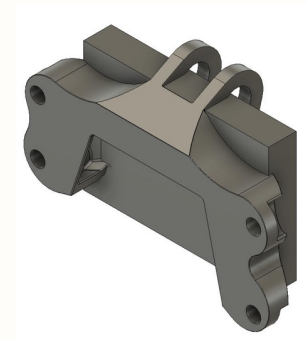
LLM 1



LLM 2



LLM 3



Improvement - Clarify intent, then guide and validate design

A Request Interpreter resolves ambiguity; ontology guidance keeps LLM-generated designs feasible.

RQ1: REQUEST INTERPRETER

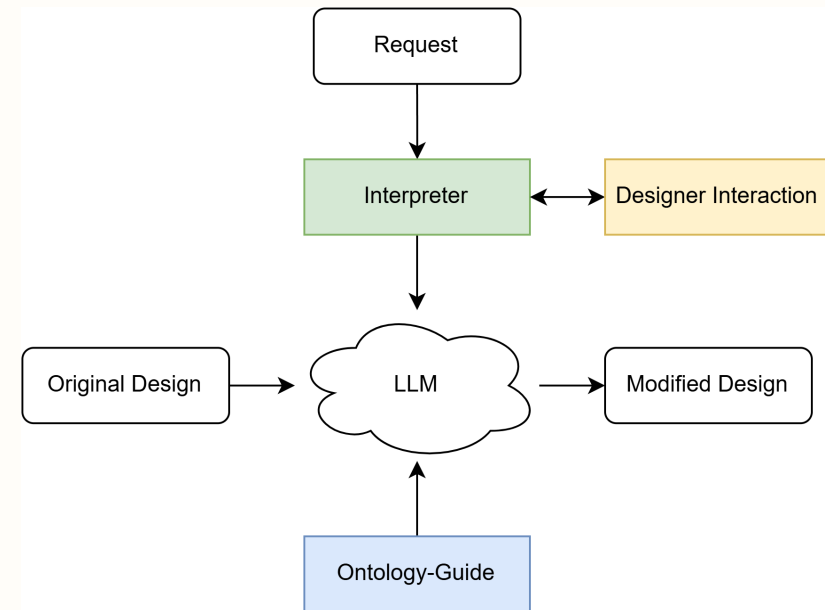
Detect whether Function, Reference Geometry, or Operation is missing or ambiguous.

If unclear → interact with the user to clarify intent.

RQ2: ONTOLOGY-GUIDED DESIGN

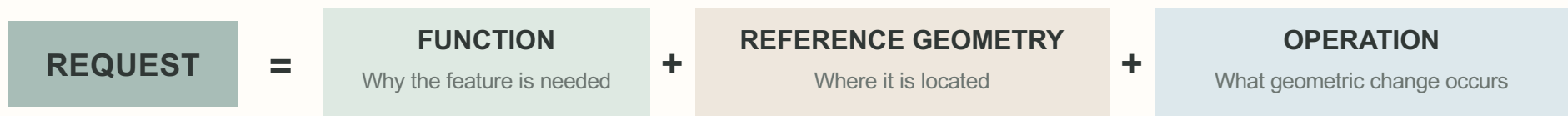
Guide LLM generation and validate functional, geometric, and structural constraints.

Outcome → diverse, feasible design alternatives.



A design request has three executable elements

Decomposing purpose, location, and action turns natural language into an executable CAD instruction.



EXAMPLE REQUEST

“Design simple locking mechanism in the area of the hook to secure carried load.”

FUNCTION

Secure the carried load

REFERENCE GEOMETRY

Hook area, especially the opening

OPERATION

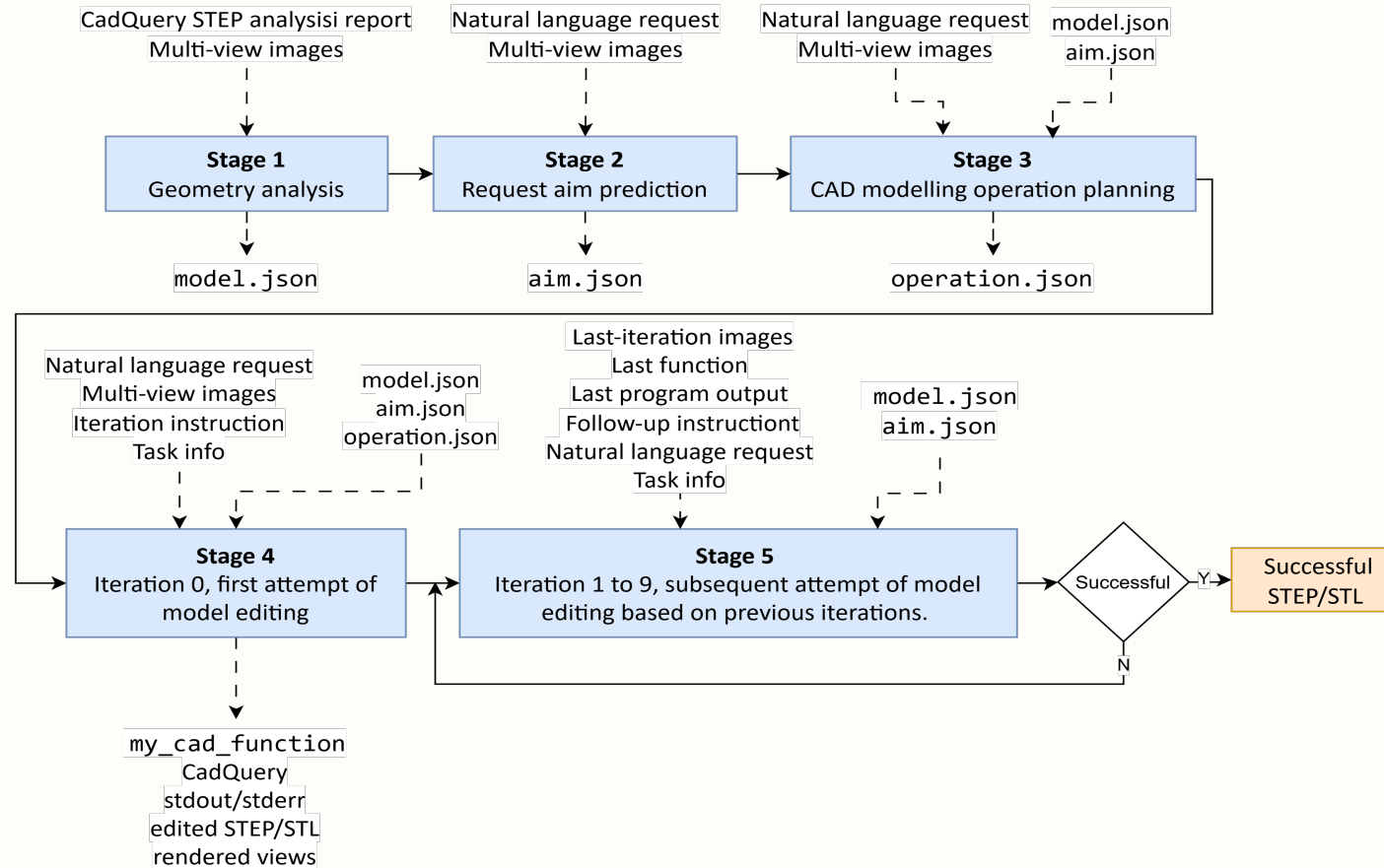
Add a simple locking latch across the hook opening

A request may contain one, two, or all three elements

Each composition provides a different level of design specificity.

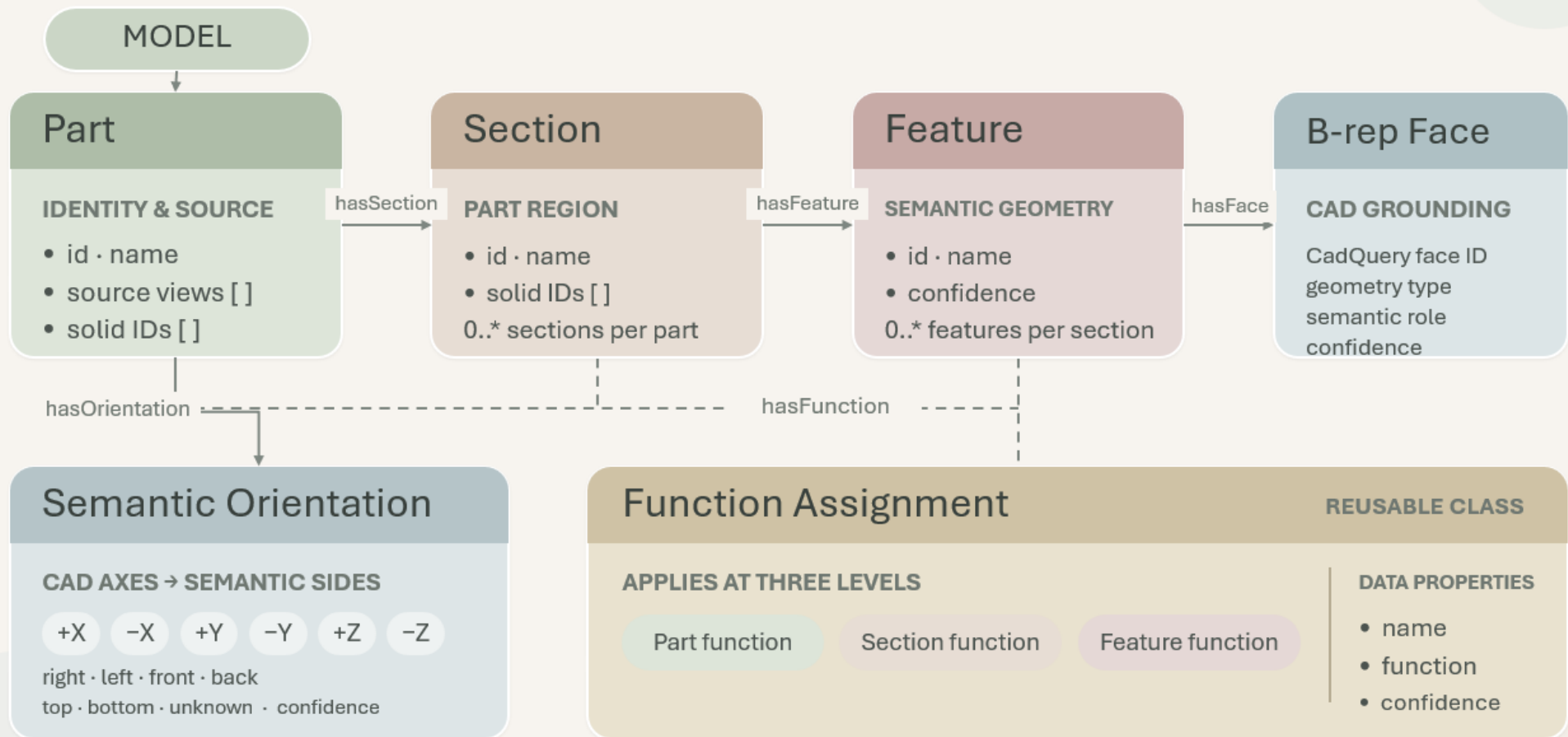
REQUEST COMPOSITION	EXPECTED DESIGN
Reference Geometry + Function	Multiple design options
Reference Geometry + Operation	Unique design
Function + Operation	Multiple design options, prone to mistakes
Reference Geometry + Function + Operation	Unique design

Pipeline Overview



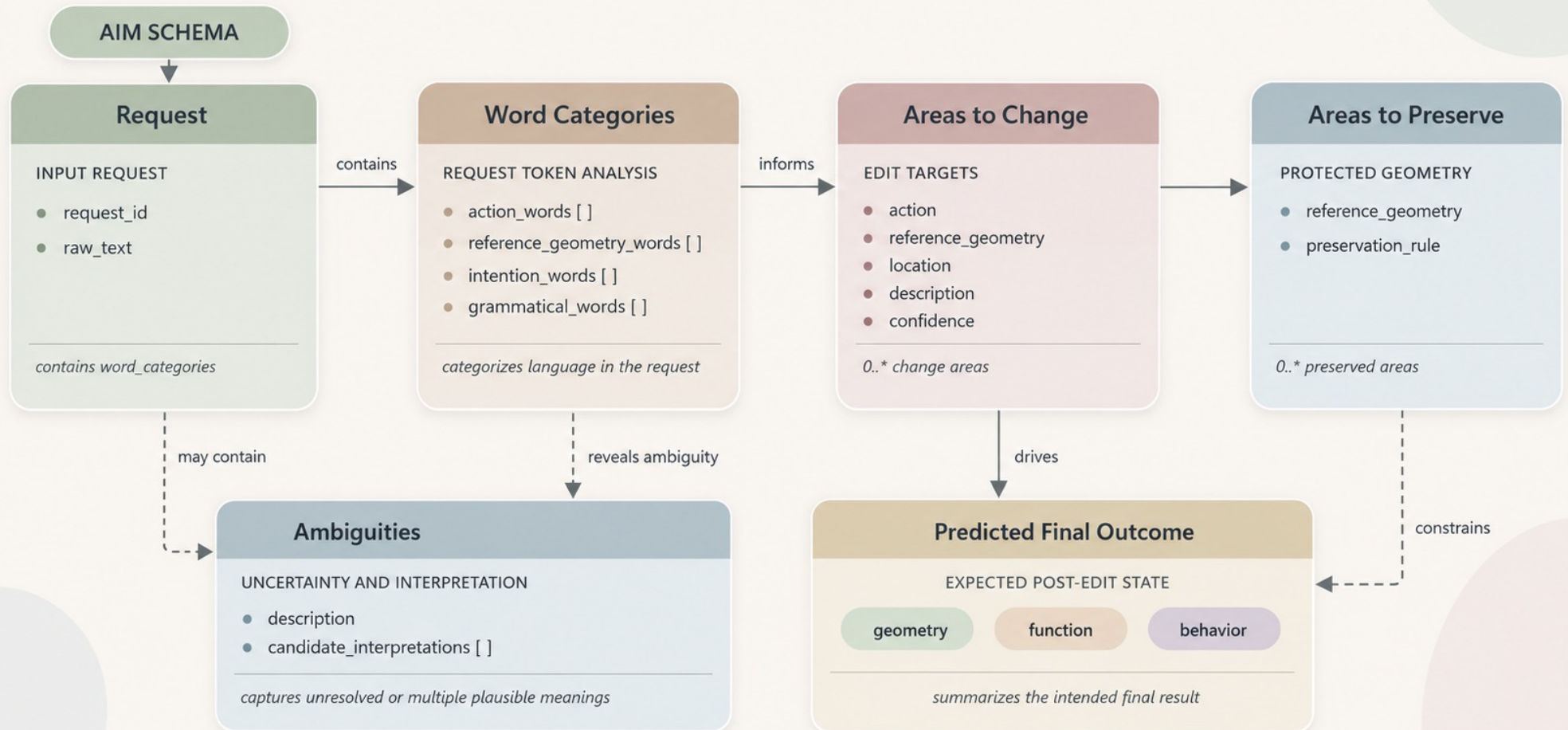
CAD Model Ontology

Semantic hierarchy and geometry grounding for model analysis



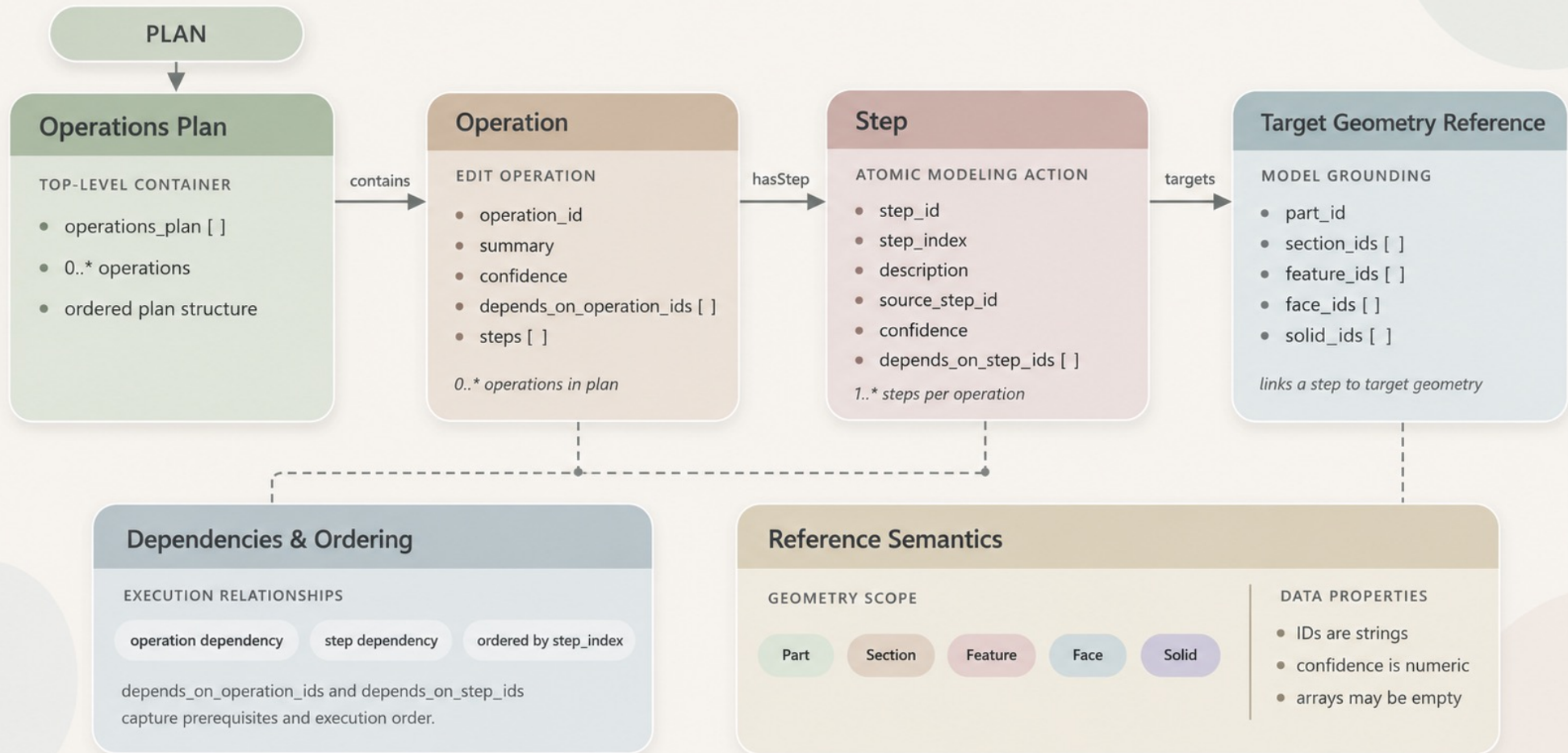
Aim Ontology

Request interpretation and edit-intent grounding for CAD modification analysis



Operation Ontology

Hierarchical planning and geometry grounding for CAD editing operations



Scores

